Module # 001	
1.	Complex Analysis related to complex numbers and their functions. We have different types of numbers which help us in different solutions. Natural Number, Integers, Rational Number i.e. $\mathbb{Q} = \{\frac{a}{b} : a, b \epsilon \mathbb{Z}, b \neq 0\}$, Irrational Numbers that cannot be written as $\frac{p}{q}$ where $p, q \epsilon \mathbb{Z}$.
2.	The decimal expression is non-terminating and non-periodic.
3.	Real Numbers: Union of rational and irrational numbers are called real numbers i.e. $\mathbb{R} = \mathbb{Q} U \mathbb{Q}'$
4.	Real numbers are not enough to solve our all problems if we consider an equation $x^2+2x+2=0$ then we get $x = \sqrt{-1}$ so we denote it i i.e. $i = \sqrt{-1}$
5.	By including i we get numbers of the form $a + ib : a, b \epsilon \mathbb{R}$ and such numbers are called complex numbers.
6.	"Cardano" was first to introduce complex numbers $a + \sqrt{-1}\, b$ into algebra but had misgivings about it.
7.	"Rafael Bombelli authored I'Algebra (1572 and 1579) a set of three books "At first , the things seemed to me to be based more on sophism than on truth, but I searched until I found the proof".
8.	L.Euler (1701-1783) introduced the notation $i = \sqrt{-1}$
Module # 002	
9.	A complex number z can be defined as an ordered pair z=(x,y) where x,y are real numbers.
10.	Complex numbers can be interpreted as points in the complex plane.
11.	Real numbers is a subset of complex numbers.
12.	When real numbers x are displayed as points (x,0) on the real axis the set of complex numbers includes the real numbers as a subset.
13.	Complex numbers of the form (0,y) corresponds to y-axis and are pure imaginary numbers when $y \neq 0$. The y-axis is then referred to the imaginary axis.
14.	For complex number z=(x,y) the real numbers x,y are known as the real and imaginary parts of z respectively and we write as x=Re z and y=Im z
15.	Two complex numbers z1 and z2 are equal whenever they have the same real parts and the same imaginary parts.
16.	The statement z1=z2 means that z1 and z2 corresponds to same point in the complex plane or z-plane.
17.	Let z1=(x1,y1), z2=(x2,y2) then z1=z2 iff x1=x2 and y1=y2.
Module # 003	
18.	Two ways to represent complex numbers are Addition of Complex numbers and Product of Complex Numbers.

19.	Commutative Law w.r.t Addition and Associative Law w.r.t Addition holds in Complex Numbers.
20.	**Addition of two complex numbers:** For z1=(x1,y1), z2=(x2,y2) the addition is defined as z1+z2=(x1,y1,)+(x2,y2,)=(x1+x2, y1+y2)
21.	**Product of two complex numbers:** For z1=(x1,y1), z2=(x2,y2) the product is defined as z1.z2=(x1,y1,).(x2,y2,)=(x1x2-y1y2, x1y2+x2y1)
22.	Any complex number z=(x,y) can be written as z=(x,y)=(x,0)+(0,y)
23.	Any complex number z=(x,y) can be written as z=(x,y)=(x,0)+(0,y), Also we have (0,1)(y,0)=(0,y) so we can write z=(x,0)+(0,1)(y,0). The numbers of the form (x,0) are real numbers so can simply denoted by x, numbers of the form (y,0) are real numbers so can simply denoted by y and (0,1) is the number denoted by "i" then z=(x,0)+(0,1)(y,0)=x+iy
24.	(0,1) is such a complex number when it is multiplied with itself then it gives -1. i.e (0,1)(0,1)=(0-1,0+0)=(-1,0)=-1or i^2=-1
25.	Z=x+iy is used to represent the complex numbers and this representation is due to the mathematician Gauss.
Module # 004	
26.	Properties of Addition of Complex Numbers: i. The addition of complex numbers follows all the properties of real numbers. ii. **Additive Identity**: There is a unique complex number 'w' such that z+w=w+z=z such number w is 0+i0 or (0,0). iii. **Additive Inverse**: For every complex number z=x+iy there is a unique ξ such that z+$\xi = 0$, obviously $\xi = -x - iy$ and is known as additive inverse.
27.	Complex numbers can be represented by two ways. i. Ordered pair representation ii. z=x+iy form. Ordered Pair Representation of complex numbers is the real foundational definition on which complex number system is built while z=x+iy is used or preferred due to simplicity.
Module # 005	
28.	Properties of Multiplication of Complex Numbers: i. The product of complex numbers follows all the properties of real numbers. ii. The commutative law z1z2=z2z1. The commutative implies iy=yi so z=x+iy and z=x+yi are same. iii. The associative law z1(z2z3)=(z1z2)z3. We can write z1z2z3 without parenthesis. iv. **Multiplicative Identity**: There is a unique complex number 'w' such that zw=wz=z such number w is 1+i0 or (1,0). iv. **Multiplicative Inverse**: For every complex number z=x+iy other than (0,0) there is a unique z^{-1} such that z$z^{-1} = 1$, In fact $z^{-1} = \frac{1}{z} = \frac{1}{x+iy}$

29.	Division of Complex numbers can be calculated using $\frac{z1}{z2} = \left(\frac{x1x2+y1y2}{x2^2+y2^2}, \frac{y1x2-x1y2}{x2^2+y2^2}\right)$
30.	**Distributive Law**: The Operation of multiplication is distributive over addition that is z1(z2+z3)=z1z2+z1z3
31.	**Binomial Formula**: For any two complex numbers z1, z2 we have $(z1+z2)^n=\sum_{k=0}^{n}\binom{n}{k} z_1^k z_2^{n-k}$ $for\ n = 1,2,3, \ldots \ldots Here\ \binom{n}{k} = \frac{n!}{k!(n-k)!}$
	Module # 006
32.	A complex number 'z' can be associated with the directed line segment or vector from origin to the point of z.
33.	Multiplication of two complex number is neither dot product nor vector product.
34.	The vector interpretation helps us defining modulus or absolute value of complex number.
35.	The modulus or absolute value of a complex number z=(x,y)=x+iy is defined and denoted as $\lvert Z\rvert = \sqrt{x^2+y^2}$
	Module # 007-008
36.	Properties of modulus of a complex number: An important consequence of triangular inequality is $\lvert Z_1 + Z_2\rvert \geq \lvert\lvert Z_1\rvert - \lvert Z_2\rvert\rvert$
37.	The complex conjugate or simple conjugate of a complex number z=x+iy is denoted by $\bar{z}$ and is given by; $\bar{z} = (x,-y) = \overline{(x+iy)} = x - iy$
38.	For two complex numbers z1 and z2 we have $\overline{z1+z2} = \overline{z1} + \overline{z2}$, $\overline{z1-z2} = \overline{z1}$ - $\overline{z2}$, $\overline{z1z2} = \overline{z1}\,\overline{z2}$ and $\overline{\left(\frac{z1}{z2}\right)} = \frac{\overline{z1}}{\overline{z2}}$ if $\overline{z2} \neq 0$
39.	Complex conjugate of z=x+iy are Re(z)=x and Im(z)=y
40.	As z=x+iy and $\bar{z}$=x-iy so $z + \bar{z} = 2x\ or\ x = \frac{z+\bar{z}}{2}\ or\ Re(z) = \frac{z+\bar{z}}{2}\ and\ z - \bar{z}$=2iy or y=$\frac{z-\bar{z}}{2i}\ or\ Im(z) = \frac{z-\bar{z}}{2i}$
41.	By following above point(in 40) we can prove that $Re(iz) = -Im(z)\ and\ Im(iz) = Re(z)$
42.	Complex conjugate helps us in proving properties of modulus . If z=x+iy then $\lvert z\rvert = \sqrt{x^2+y^2}$
43.	$\lvert z\rvert^2 = z\bar{z}$
44.	Complex conjugate fulfill triangular inequality property i.e. $\lvert z_1 + z_2\rvert \leq \lvert z_1\rvert + \lvert z_2\rvert$
	Module # 009
45.	The number z=x+iy can be written in polar form as z=(rCos$\theta, rSin\theta)\ or\ z = r(Cos\theta + iSin\theta)$ and the values of r and θ are known as polar coordinates of Z.
46.	If z=0 then coordinates of θ are undefined.
47.	The length of $r = \lvert z\rvert$ is not allowed to be negative since it is length of radius vector for z.

48.	The real number θ represents the angle measured in radians that z makes with positive x-axis.
49.	Polar representation of z=1+i is $z=\sqrt{2}\left(Cos\left(\frac{\pi}{4}\right)+Sin\left(\frac{\pi}{4}\right)\right)$**(PREPARE EXAMPLE P#25)**
50.	For a complex number z=(x,y) the coordinates θ has infinite number of values. All these values differ by integral multiples of 2π
51.	**Define Arguments of Z=(x,y)** : If $z\neq 0$, arg(z)={ θ :z=r(Cos θ +i Sin θ)}
52.	If $z\neq 0$ be a complex number then principal value of argument of z is denoted by Arg z is a unique value of θ such that z=r(Cos θ +i Sin θ) and $-\pi<\theta\leq\pi$
53.	Principal value of Arg z is also referred as "The argument of z" so we can say that arg z=Arg z +2 $n\pi$ where n is any integer.
54.	If x=0 then we can not use $tan\theta=\left(\frac{y}{x}\right)$ to find arg z.
55.	If $z\neq 0$ and x=0 then we use the following rule Arg z=$\frac{\pi}{2}$ if Im z>0 and Arg z=-$\frac{\pi}{2}$ if Im z<0
Module # 010	
56.	The expression $e^{i\theta}$ *or* $\exp(i\theta)$ is defined by means of Euler formula as $e^{i\theta}=Cos\ \theta+i\ Sin\theta$
57.	The exponential of complex number in Euler formula is $z=re^{i\theta}$
58.	Euler's Identity: For $\theta=\pi$ we get z=$e^{i\pi}=Cos\ \pi+iSin\pi=-1$ so we have $e^{i\pi}+1=0$ this is known as Euler's Identity.
Module # 011	
59.	The expression $(Cos\ \theta+i\ Sin\theta)^n=Cos\ n\theta+iSinn\theta$ where n=0,$\pm$1, $\pm$2, $\pm$3, $\pm$4, ... is known as De-Moivre's Formula.
Module # 012-013	
60.	A number u is said to be an nth root of complex number z if u^n=z
61.	If p(z) is a polynomial of degree n (n>0) with complex coefficient then equation p(z)=0 has precisely n solutions.
62.	There are precisely n number of nth roots of a complex number z_0.
63.	We denote the set of all nth roots of a complex number z0 by $z_0^{\frac{1}{2}}$
64.	For positive real number r_0 the symbol $\sqrt[n]{r_0}$ is reserved for one positive root.
65.	Equality of complex number in exponential form is $z=re^{i\theta}$

66.	Consider a complex number $z_0 = r_0 e^{i\theta_0}$ then the nth root of z is given by $z = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)\right], k = 9, \pm1, \pm2, \pm3, \pm4, ----$
Module # 014-016	
67.	Equation of a circle of radius ε and center z_0 is given by $\lvert z - z_0 \rvert = \varepsilon$
68.	A closed disk with center z_0 and radius ε is defined by $\lvert z - z_0 \rvert \leq \varepsilon$
69.	A open disk with center z_0 and radius ε is defined by $\lvert z - z_0 \rvert < \varepsilon$
70.	A point z_0 is **interior point** of a set S whenever there is a neighborhood of z_0 contained in S.
71.	A point z_0 is **exterior point** of a set S whenever there is a neighborhood of z_0 not contained in S.
72.	A point z_0 is **boundary point** of a set S if it is neither an interior point nor an exterior point.
73.	A set S is called an **Open Set** if every point of S in an interior point of S.
74.	A set S is called a **Closed Set** if it contains all its boundary points.
75.	A point z_0 is called an **accumulation Point** of a set S is each deleted neighborhood of z_0 contains at least one point of S.
76.	If a set is closed then it contains all its accumulation points.
77.	**Closure of a set** S is the closed set consisting of all points in S together with boundary points.
78.	Some sets are neither open nor closed.
79.	Some sets are open as well as closed.
80.	A curve is piece of string placed on a flat surface.
81.	A curve C is the range of a function given by z(t)=(x(t),y(t))=x(t)+iy(t) where t∈ $[a, b]$
82.	A curve is called **simple** if it does not cross itself.
83.	A set S is said to be **connected set** if for every pair of points $z1, z2 \in S$ there is a curve C joining them which is entirely contained in S.
84.	A connected open set is called a domain.
85.	A set is said to be **bounded set** if it can be completely contained in some closed disk.
86.	A set which is not bounded is called unbounded.
Module # 29	

87.	Continuity means without any break or sudden jump or holes at any point.
88.	If the small change in the input of a function brings to a small change in the output of that function then function is continuous
89.	A function f(x) is continuous at a point X_0 if • $\lim_{x\to x_0} f(x)$ Exists. • $f(x_0)$ Exists. • $\lim_{x\to x_0} f(x) = f(x_0)$
90.	A function is said to be continuous on a set A if it is continuous at each point of A.
91.	The mean value theorem for complex numbers does not hold.
92.	In order to calculate limit at a point x_0 function f(x) must be defined in neighborhood of x_0.
93.	The function $f(x) = \begin{cases} x^2+1 & x\leq 0 \\ x & x>0 \end{cases}$ is continuous at x=1 but not continuous at x=0
94.	Let U(x,y) be a real valued function of the two real variables x and y then U is continuous at a point (x_0, y_0) if the following three conditions are satisfied; • $\lim_{(x,y)\to(x_0,y_0)} U(x,y)$ Exists. • $U(x_0,y_0)$ Exists. • $\lim_{(x,y)\to(x_0,y_0)} U(x,y) = U(x_0,y_0)$
95.	The function $U(x,y) = \begin{cases} \frac{x^3}{x^2+y^2} = 0 & (x,y)\neq(0,0) \\ 0 & (x,y)\neq(0,0) \end{cases}$ is continuous at (0,0)
96.	A complex function f(z) of complex variable z for all values of z in some neighborhood of z_0 is continuous at z_0 if it satisfied the following three conditions; • $\lim_{z\to z_0} f(z)$ Exists. • $f(z_0)$ Exists. • $\lim_{z\to z_0} f(z) = f(z_0)$
97.	For every $\varepsilon > 0$ *there exist a* $\delta > 0$ such that $\lvert f(z) - L\rvert < \varepsilon$ whenever $0 < \lvert z - z_0\rvert < \delta$ Where $\lim_{z\to z_0} f(z) = f(z_0) \quad \Longrightarrow \quad \lim_{z\to z_0} f(z) = L$
98.	A complex function f(z) of complex variable z for all values of z in some neighborhood of z_0 is continuous at z_0 if for each $\varepsilon > 0$ *there exist a* $\delta > 0$ such that $\lvert f(z) - f(z_0)\rvert < \varepsilon$ whenever $0 < \lvert z - z_0\rvert < \delta$
	Module # 30
99.	Let a function f(z) = u(x, y) + i v(x, y) be defined in some neighborhood of z_0 then f is continuous at $z_0 = x_0 + iy_0$ iff u and v are continuous at (x_0, y_0)

100.	If f and g are continuous at the point z_0 then the following functions are continuous at z_0. • The sum f+g, where (f+g)(z)=f(z)+g(z) • The difference f-g, where (f-g)(z)=f(z)-g(z) • The Product fg, where (fg)(z)=f(z)g(z) • The quotient $\frac{f}{g}$, where $\left(\frac{f}{g}\right)(z) = \frac{f(z)}{g(z)}$ provided $g(z) \neq 0$
101.	Any polynomial $\mathrm{p(z) = a_0 + a_1 z + a_2 z^2 + \cdots \ldots \ldots .. + a_n z^n (a_n \neq 0)}$ is continuous at each point $\mathrm{Z_0}$ in the complex plane.
102.	$\lim_{z\to i} \frac{z^2+2}{z^2-2z+1} = \frac{1}{-2i}$
Module # 31	
103.	Consider two complex valued functions f(z) and g(z) then its composition is denoted by '**g of (z)**' and is defined as '**gof(z)=g(f(z))**'
104.	If f and g are continuous at the point z_0 then 'gof' is continuous at z_0.
Module # 32	
105.	If a function f(z) is continuous and non zero at a point z_0 then f(z) $\neq$ 0 throughout some neighborhood of that point.
106.	If a function f is continuous at throughout a region R that is both closed and bounded then there exist a non real number M such that $\lvert f(z) \leq M \rvert$ for all points Z in R where equality holds for at least one such Z.
Module # 33	
107.	The mapping $\frac{1}{Z}$ is called the reciprocal function. In exponential notation it can be expressed as $z = re^{i\theta}$ then we obtain $w = \frac{1}{Z} = \frac{1}{re^{i\theta}} = \frac{1}{r}e^{-i\theta}$ Case 1: If z lies on a unit circle. Case 2: If z lies outside the unit circle. Case 3: If z lies inside the unit circle.
108.	Image of the half plane {z:Re(z)≥ $\frac{1}{2}$ under the mapping $w = \frac{1}{Z}$ is the closed disk i.e. {w:|w-1|≤ 1}
109.	For transformation $\frac{1}{Z}$ image of portion of the half plane Re(z) ≥ $\frac{1}{2}$ is inside the closed disk i.e. {z:|z-$\frac{1}{2}$|≤ 1}
Module # 34	
110.	The derivative of 'f' at x_0 is written as $f'(x_0)$ and is defined as $f'(x_0) = \lim_{\mathrm{x\to x_0}} \frac{\mathrm{f(x)-f(x_0)}}{\mathrm{x-x_0}}$ provided the limit exists or $f'(x_0) = \lim_{\Delta_x\to 0} \frac{f(x_0+\Delta_x)-f(x_0)}{\Delta_x}$
111.	The derivative of f at z_0 is written as $f'(\mathrm{z_0}) = \lim_{\mathrm{z\to z_0}} \frac{\mathrm{f(z)-f(z_0)}}{\mathrm{z-z_0}}$ provide the limit exists. The function f must be defined is a neighborhood of z_0

112.	If we write $\Delta_z = z - z_0$, *then the funciton* $f'(z_0) = \lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0}$ can be expressed as $f'(z_0) = \lim_{\Delta_z\to 0} \frac{f(z_0+\Delta_z)-f(z_0)}{\Delta_z}$
113.	If we let w=f(z) and $\Delta w = f(z) - f(z_0)$ then we can use the Leibniz notation $\frac{dw}{dz}$ for the derivative $f'(z_0) = \frac{dw}{dz} = \lim_{\Delta_z\to 0}\frac{\Delta w}{\Delta z}$ Hence $f'(z_0) = \lim_{\Delta_z\to 0}\frac{\Delta w}{\Delta z} = \lim_{\Delta_z\to 0}\frac{f(z_0+\Delta_z)-f(z_0)}{\Delta_z}$ and often we drop the subscript z_0 and write the definition as $f'(z) = \lim_{\Delta_z\to 0}\frac{\Delta w}{\Delta z} = \lim_{\Delta_z\to 0}\frac{f(z+\Delta_z)-f(z)}{\Delta_z}$
114.	If $f(z) = z^2$ then $f'(z) = 2z$. **(PROVE IT-PAGE 66)**
115.	The function f(z)=Re(z) is differentiable nowhere. **(PROVE IT-PAGE 67)**
	Module # 35
116.	Consider the real valued function f(z)=$\lvert z\rvert^2$. Find the points where f(z) is differentiable. **(PROVE IT-PAGE 69)**
117.	If f(z) is differentiable at z_0 then f(z) is continuous at z_0. **(PROVE IT-PAGE 73)**
	Module # 36
118.	Chain rule is useful when we want to differentiate composition of two functions.
119.	Given two functions f(z) and g(z) composition of f and g is given by; fog(z)=f(g(z)) or $\frac{d}{dz}(f(g(z)) = f'(g(z)).g'(z))$ **(PREPARE EXAMPLE P#78)**

	Module # 37 – The Cauchy – Riemann Equation
120.	Augustin-Lonis Cauchy (French Mathematician, Engineer and Physicist – 21 March 1789 to 23 May 1857) and Bernhard Riemann.
121.	Cauchy – Riemann equations are a pair of equations that first order partial derivative of the component function u and v of a function f(z)=u(x,y)+iv(x,y) must satisfy at a point (x_0,y_0)when derivative of f exist there.
122.	If Cauchy – Riemann equations are not satisfied at z_0 then function is not differentiable at z_0.
123.	A Cauchy – Riemann equations are satisfied at a point z_0 then function may or may not be differentiable at z_0. **(PROVE IT-PAGE 81)**
	Module # 38
124.	If f(z)=f(x+iy)=u(x,y)+iv(x,y) differentiable at a point z_0=x_0+iy_0 then u and v satisfy Cauchy-Riemann equations Ux(x_0,y_0)=Vy(x_0,y_0) and Uy(x_0,y_0)=-Vx(x_0,y_0)
	Module # 39
125.	Let (x_0,y_0) be an arbitrary point defined in some neighborhood of the point Z_0 =x_0+iy_0 and suppose that f(z)=f(x+iy)=u(x,y)+iy(x,y) then U(x,y)=e^{-y}Cosx ------- Ux(x,y)=-e^{-y}Sinx -------Uy(x,y)= -e^{-y}Cosx and V(x,y)=e^{-y}Sinx ------- Vx(x,y)=-e^{-y}Cosx -------Vy(x,y)= -e^{-y}Sinx

126.	The above partial derivatives are continuous at (X0,Y0) and satisfy Cauchy – Riemann equations Ux=Vy and Uy=-Vx i.e. Ux=)=-e^{-y}Sinx =Vy and Uy=-e^{-y}Cosx =-Vx
Module # 41	
127.	A complex function f is analytic at the point Z_0 provided there is some $\varepsilon > 0$ such that $f^{'}(z)$ exist in ε neighborhood of Z_0.
Module # 45	
128.	Harmonic Functions are real valued functions but they are very strongly related to complex valued functions.
129.	**<u>Harmonic Function Definition</u>**: A real valued function H of two variables x and y is said to be harmonic in a given domain D of xy-plane if following two conditions are satisfied by function; 1. Throughout the domain it has continuous partial derivatives of the first order and second order. 2. It satisfied following partial differential equation Hxx(x,y)+Hyy(x,y)=0 or $\frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2} = 0$
130.	Notations: The sum $\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2}$ of two partial derivatives is denoted by $\nabla^2 \varphi$ and is called the Laplacian of φ.
131.	Laplacian equation or sometimes known as Potential equation is $\nabla^2 \varphi = 0$ **(PREPARE EXAMPLE - PAGE 27)**
Module # 46	
132.	If a function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D, then its component functions u and v are harmonic in D. **(PREPARE EXAMPLE - PAGE 28)**
133.	**<u>Harmonic Conjugate</u>**: Let $u(x, y)$ and $v(x, y)$ are two functions such that $u(x, y)$ and $v(x, y)$ are harmonic in D and first partial derivatives of U and V satisfy Cauchy-Riemann equations throughout D then V is said to be harmonic conjugate of U in D.
134.	A function f(z)=U(x,y)+iv(x,y) is analytic in a domain D if and only iff V is harmonic conjugate of U.
135.	The function $f(z)=z^2$ i.e $f(x+iy)=(x+iy)^2=(x^2-y^2)+i(2xy)$ is analytic everywhere. It means $U(x,y)= x^2-y^2$ and $V(x,y)= 2xy$
136.	The function 2xy and x^2-y^2 are harmonic functions but $f(x+iy)=(x+iy)^2=(2xy)+i(x^2-y^2)$ is not harmonic.
Module # 47	
137.	Find Harmonic Conjugate of $U(x,y)=x^3-3xy^2-5y$. **(PREPARE EXAMPLE - PAGE 32)**
Module # 48	
138.	Harmonic Functions are very important in applied mathematics for solving real life problems e.g. for fluid flow problems.
139.	Incompressible means density w.r.t times remains constant or density does not change with time and Frictionless means when fluid flows then its friction with the walls is zero.
140.	An analytic function has analytic anti-derivative.

141.	The curves given by {(x,y):$\varphi(x, y) = constant$} are called equi-potentials or stream lines.
142.	For F(x,y)= $\varphi(x, y) + i\Psi(x, y)$ the component $\varphi(x, y)$ is harmonic also $\varphi_y = -\Psi_x$
	Module # 49
143.	**Sequence:** A sequence is any collection of objects, events or numbers that has some pattern that we can find e.g. 1,2,3,5,8,13,…….
144.	A sequence is a function whose domain is a positive integer and whose range is a subset of complex numbers.
145.	$f: z^+ \rightarrow \mathbb{C}$ or $f: \mathbb{N} \rightarrow \mathbb{C}$ is a sequence
146.	Examples of complex valued sequence: $f_1(n) = n + i2n, \quad n \in \mathbb{N}$
147.	A sequence is usually denoted as $\{Z_n\}_{n=1}^{\infty} \ OR \ \{Z_n\}_1^{\infty}$
	Module # 50
148.	A sequence is function from natural number to complex number.
149.	**Limit of a sequence**: A sequence {Xn} has P as its limit as n approaches to infinity provided the terms Xn can be made as close as we want to P by making n large enough.
150.	A sequence of real numbers {Xn} converges of P if for every $\varepsilon > 0$ there exist Nε such that |Xn-P|< ε whenever n>N
151.	A sequence of complex numbers {Zn} converges to ζ if for each $\varepsilon > 0$ there exist Nε scuh that |Zn- ζ|< ε whenever n>Nε denotes as $\lim_{n\to\infty} Zn = \zeta$

	Module # 51
152.	A complex sequence {Zn} is bounded provided that there exist a positive real number R and an integer N such that |Zn|<R for all n>N.
153.	If {Zn} is a convergent sequence then {Zn} is bounded.
154.	**Cauchy Sequence**: The sequence {Zn} is a Cauchy Sequence if for every $\varepsilon > 0$ there is a positive integer Nε such that if n,m>Nε then |Zn-Zm|<ε
155.	{Zn} is a Cauchy Sequence if and only if {Zn} converges.

S#	Statement	Answer
156.	The derivative of f at z_0 i.e. $f' = \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}$ provided the limit exists. The function must be defined is a neighborhood of ___	z_0
157.	If Cauchy – Riemann equations are not satisfied at a point z_0 then function is not _______ at z_0.	Differentiable
158.	In the complex valued function z=x-iy, the value of Ux=___________	1
159.	In the complex valued function z=x-iy, the value of Vx=___________	0
160.	In the complex valued function z=x+iy, the value of Ux=___________	1
161.	In the complex valued function z=x+iy, the value of Vx= _________	0
162.	Consider $\lim_{z\to z_0} f(z) = A$ and $\lim_{z\to z_0} g(z) = B$ then the $\lim_{z\to z_0}(f(z) \pm g(z)) =$ _______	$A \pm B$
163.	Consider $\lim_{z\to z_0} f(z) = A$ and $\lim_{z\to z_0} g(z) = B$ then the $\lim_{z\to z_1}(f(z)g(z)) =$ _______	A.B
164.	If Cauchy – Riemann on a pair of real – valued functions of two real variables u(x,y) and v(x,y) are Ux=Vy and ___.	Uy= -Vx
165.	Each simple closed curve C divides the plane into two domains (Connected Open Sets). One of which is bounded and is called the ___ of C.	Interior
166.	Each simple closed curve C divides the plane into two domains (Connected Open Sets). One of which is unbounded and is called the _____ of C.	Exterior
167.	If we write $\Delta_z = z - z_0$, *then the funciton* $f'(z_0) = \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}$ can be expressed as $f'(z_0) =$ ___	$\lim_{\Delta_z\to 0}\frac{f(z_0+\Delta_z)-f(z_0)}{\Delta_z}$
168.	A Cauchy – Riemann equations are satisfied at a point z_0 then function ______ differentiable at z_0.	C. Both (a) and (b)
169.	Consider $\lim_{z\to z_0} f(z) = A$ and $\lim_{z\to z_0} g(z) = B$ then the $\lim_{z\to z_1}\frac{f(z)}{g(z)} =$ _______	$\frac{A}{B}, B \neq 0$
170.	If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are ____ in D.	Harmonic
171.	A curve C is said to be a contour if it is constructed by joining finitely many smooth curves end to end	True
172.	The sequence $z_n = -3 + i\frac{1}{n^3}(n = 1,2,3,4, \dots\dots)$ converges to ____________-	-3
173.	Any polynomial $p(z) = a_0 + a_1z + a_2z^2 + \cdots \dots \dots .. + a_nz^n (a_n \neq 0)$ of degree $n(n \geq 1)$ has least one zero, i.e. there exists at least ------------- Z_0 such that $P(Z_0) = 0$	One point
174.	The sequence $z_n = \frac{1}{n^3}(n = 1,2,3,4, \dots \dots)$for $n \to \infty$ converges to _______.	0

175.	A domain D (Connected Open Set) is called __________ connected if interior of every simple closed contour C is contained in D.	Simply
176.	If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to_____.	c=1 and a=-b
177.	A function V is said to be harmonic if and only if ________________	Vxx+Vyy=0
178.	A function f(z) is analytic function if ____________ part of f(z) is analytic.	C. Both (a) and (b)
179.	Harmonic conjugate of $u(x,y)=e^y \cos x$ is ________.	$-e^y \sin x + C$
180.	If $e^{ax}\cos y$ is harmonic then 'a' is ___________.	i
181.	If U and V are harmonic functions then f(Z)=U+iV is______________.	Analytic Function
182.	A point at which a function ceases to be analytic is called a ________ point.	Singular
183.	Cauchy-Riemann equations are	Ux=Vy and Uy=-Vx
184.	If f(z)=u+iv in polar form is analytic then $\frac{\partial u}{\partial r}$ is	$\frac{1}{r}\frac{\partial v}{\partial \theta}$
185.	If f(z)=u+iv in polar form is analytic then $\frac{\partial u}{\partial \theta}$ is	$-r\frac{\partial v}{\partial r}$
186.	A function v is called a conjugate harmonic function for a harmonic function u in Ω whenever	f=u+iv is analytic
187.	The function $f(z) = \|z\|$ is non-constant.	nowhere analytic function
188.	The function $f(x+iy)=x^3+ax^2y+bxy^2+cy^3$ is analytic if	a=3i, b= -3 and c=i
189.	There exist no analytic function f such that	Re $f(z)=y^2-2x$
190.	The harmonic conjugate of $2x-x^3+3xy^2$ is	$2y-3x^2y+y^3$
191.	The harmonic conjugate of u(x,y)=2x(1-y) is	x^2-y^2+2y+C
192.	If the real part of an analytic function f(z) is x^2-y^2-y, then imaginary part is	2xy+x
193.	If the imaginary part of an analytic function f(z) is 2xy+y, then real part is	x^2-y^2+x
194.	$f(z) = \bar{z}$ is differentiable	Nowhere
195.	$f(z) = \|\bar{z}\|^2$ is differentiable	Only at z=0
196.	$f(z) = \|\bar{z}\|^2$ is	Differentiable at z=0 but not

		analytic at z=0
197.	$if\ f(z) = \begin{cases} \frac{xy}{(x^2+y^2)}, & if\ \ z \neq 0 \\ 0 & if\ z = 0 \end{cases}$ then f(z) is	Not differentiable at z=0
198.	$f(z) = e^z$ is analytic	Everywhere
199.	$e^x(\cos y - i \sin y)\ is$	Not analytic
200.	If f(z) is analytic then $\overline{f(\bar{z})}$ is	Analytic
201.	The point at which $f(z) = \frac{z^2-z}{(z^2-3z+2)}$ is not analytic are;	*i and -i*
202.	The harmonic conjugate of u=log$\sqrt{x^2+y^2}$ is	$tan^{-1}\left(\frac{y}{x}\right)$
203.	If $f(z) = z(2-z)$ then f(1+i)=	i
204.	If $f(z) = \lvert\bar{z}\rvert$ then f(3-4i)=	5
205.	Critical point of bilinear transformation $w = \frac{a+bz}{c+dz}$ are	$-\frac{c}{d}, \infty$
206.	The point coincide with their transformation are known as	Fixed points
207.	$w = \frac{a+bz}{c+dz}$ is a bilinear transformation when	$ad - bc \neq 0$
208.	$w = \frac{1}{z}$ is known as ________--	Inversion
209.	$w = z + \alpha$ is known as __________-	Translation
210.	A translation of the type $w = \alpha z + \beta$ where $\alpha\ and\ \beta$ are the complex constants, known as a ____	Linear transformation
211.	A mapping that preserves angles between oriented curves both in magnitude and in sense is called a/an _____ mapping.	Conformal
212.	The mapping defined by an analytic function $f(z)$ is conformal at all points z except at points where _____	$f'(z) = 0$
213.	The fixed points of the transformation $w = z^2$ are ________	0,1
214.	The invariant points of the mapping $w = \frac{z}{2-z}$ are _______	0,1
215.	The fixed points of $w = \frac{z-1}{z+1}$ are ______	$\pm i$
216.	The mapping $w = z + \frac{1}{z}$ transforms circles of constant radius into________.	confocal ellipses

217.	Under the transformations $w = \frac{1}{z}$,the image of the line $y = \frac{1}{4}$ in z-plane is ____	Circle $u^2 + v^2 + 4v = 0$
218.	The bilinear transformation that maps the points $0, i, \infty$ respectively into $0,1,\infty$ is $w =$ _______	$-iz$
219.	The bilinear transformation which maps the points $z = 1, z = 0, z = -1$ of z-plane into $w = i, w = 0, w = -1$ of w-plane respectively is ____	$w = iz$

MODULE#055 to85	
220.	**Geometric Series**: A finite series of the form $\sum_{n=0}^{\infty} Z^n$ is called geometric series.
221.	For what value of Z the geometric series $\sum_{n=0}^{\infty} Z^n$ converges? If $\lvert z\rvert < 1$ the series $\sum_{n=0}^{\infty} Z^n$ converges to $f(z) = \frac{1}{(1-z)}$. If If $\lvert z\rvert \geq 1$ the series $\sum_{n=0}^{\infty} Z^n$ diverges. **PROOF M#55-P#2**
222.	If $\lvert z\rvert > 1$ the series $\sum_{n=1}^{\infty} Z^{-n}$ converges to $f(z) = \frac{1}{(1-z)}$. If If $\lvert z\rvert \geq 1$ the series $\sum_{n=0}^{\infty} Z^n$ diverges.
223.	If $\sum_{n=0}^{\infty} \zeta_n$ is a complex series such that $\lim_{n\to\infty} \frac{\lvert\zeta_{n+1}\rvert}{\lvert\zeta_n\rvert} = L$ then the series is absolutely convergent if $L < 1$ and divergent if $L > 1$
224.	**PREPARE EXAMPLE....M#56-P#5**
225.	**Accumulation Point:** A point $Z_0$is called an accumulation point of a set S is each deleted neighborhood of Z_0 contains at least one point of S.
226.	**Limit Supermum**: Let $\{t_n\}$ be a sequence of positive real numbers. The limit supremum of the sequence is denoted by $\lim_{n\to\infty} Sup\ t_n$ is the smallest real number L having the property that for any $\varepsilon > 0$ there are at most finitely many terms in a sequence that are larger than $L + \varepsilon$. If there is no such L then $\lim_{n\to\infty} Sup\ t_n = \infty$
227.	The series $\sum_{n=0}^{\infty} \zeta_n$ has $\lim_{n\to\infty} Sup\ \lvert\zeta_n\rvert^{\frac{1}{n}} = L$ then the series absolutely convergent if L<1 and divergent if L>1.
228.	**PREPARE EXAMPLE....M#58-P#14-15**
229.	Define a function: An infinite series or an expression of the form $\sum_{n=1}^{\infty} a_n$ is called function.
230.	The series $\sum_{n=0}^{\infty} C_n(z-\alpha)^n$ defines a function only if it converges.
231.	$f(z) = \sum_{n=0}^{\infty} C_n(z-\alpha)^n$ is called power series function.
232.	Domain: Collection of all points z for which the series converges.

233.	Suppose $f(z) = \sum_{n=0}^{\infty} C_n(z-\alpha)^n$ then the set of points z for which the series converges is one of the following : 1. The single point $z = \alpha$ 2. The disk $D_s(\alpha) = \{z: \|z-\alpha\| < p\}$ 3. The entire complex plane
234.	**Radius of convergence**: The number p such that the series $\sum_{n=0}^{\infty} C_n(z-\alpha)^n$ converges for all z satisfying $\|z-\alpha\| < p$ and diverges for all z satisfying $\|z-\alpha\| > p$ is called radius of convergence.
235.	If the series $\sum_{n=0}^{\infty} C_n(z-\alpha)^n$ converges for the single point $z = \alpha^n$ then radius of convergence=0.
236.	If the series $\sum_{n=0}^{\infty} C_n(z-\alpha)^n$ converges for the disk $Dp = \alpha^c = \{z: \|z-\alpha\| < p\}^n$ then radius of convergence=p.
237.	If the series $\sum_{n=0}^{\infty} C_n(z-\alpha)^n$ converges for the entire plane then radius of convergence=∞.
238.	**PREPARE EXAMPLE....M#61-P#25-26-27-28**
239.	$\sum_{n=0}^{\infty}(n+1)z^n = \frac{1}{(1-z)^2}$ for all $z \in D_1(0)$
240.	The function e^z is an entire function satisfying the following conditions: $\frac{d}{dz}e^z = e^z,\ e^{z1+z2} = e^{z1}.e^{z2}$
241.	Euler Formula: If θ is a real number then $e^{i\theta} = cos\theta + i\ sin\theta$.
242.	The exponential function $\exp(z) = e^x(Cosy + i\ siny)$ is not one-to-one mapping but onto mapping exist.
243.	Log(z) is a single value function from non zero complex numbers to the fundamental period strip.
244.	**PREPARE EXAMPLE....M#__P#56**
245.	$Log(z) = ln\|z\| + i(Arg(z) + 2n\pi)$ is multi valued function but $Log(z) = ln\|z\| + iArg(z)$ is a single valued function.
246.	$Log(z)$ has two properties: Continuity and differentiability .
247.	$Log(z)$ has become ln(x) when restricted to real numbers.
248.	The image of $Log(z)$ is one sheet of complex numbers.
249.	The graph of $Log(z)$ looks like many sheets one for each 'n'
250.	The function $Log(z)$ is discontinuous at all the points of negative real axis.

251.	Let $z_1 and\ z_2$ be non-complex numbers. The multi valued function log obeys the following properties. i. $\text{Log}(z_1 z_2) = \log(z_1) + \log(z_2)$ ii. $\text{Log}(\frac{z_1}{z_2}) = \log(z_1) - \log(z_2)$	
252.	The identity $log(z_1 z_2) = \log(z_1) + \log(z_2)$ hold true if and only iff $-\pi < Arg(z_1) < Arg(z_2) \leq \pi$	
253.	**PREPARE EXAMPLE….M#__P#69**	
254.	**PREPARE EXAMPLE….M#__P#74**	
255.	Find the derivative of b^2. **PREPARE EXAMPLE….M#__P#77**	
256.	If z is a complex number then $e^{iz} = cosz + i\ sinz$.	
257.	For $z = x + iy$ we have $Cos\ z = \text{Cos}\, x\ Coshy - i\ Sinx\ Sinhy$, $Sin\ (z + 2\pi) = Sin\ z$, $Cos\ (z + 2\pi) = Cos\ z$, $Sin\ (z + \pi) = -Sin\ z$, $Cos\ (z + \pi) = -Cos\ z$, $Tan\ (z + \pi) = Tan\ z$, $Cot\ (z + \pi) = Cot\ z$,	
258.	$Sin\ z = 0$ iff $z = n\pi, n \in z$	
259.	$Cos\ z = 0$ iff $z = \left(\frac{n+1}{2}\right)\pi, n \in z$	
260.	**PREPARE EXAMPLE….M#__P#92**	
261.	The periodic nature of $Sinz$ makes it a multiple one-to-one function.	
MODULE#086 to100		
262.	The parametric representation $z(t) = cost + isint$ for $0 \leq t \leq 2\pi$ represents a ________ in the complex plane.	Circle
263.	The parametric representation $z(t) = t^2 + it$ for $-1 \leq t \leq 1$ represents a ________ in the complex plane.	Parabola
264.	A curve $c: z(t) = x(t) + iy(t)$ for $a \leq t \leq b$, is simple if it does not cross it self, that is ________	$z(t_1) \neq z(t_2)$ whenever $t_1 \neq t_2$
265.	A curve $c: z(t) = x(t) + iy(t)$ for $a \leq t \leq b$, is closed if ________	$z(a) = z(b)$
266.	The complex valued function $f(t) = u(t) + iv(t)$ is said to be differentiable $[a, b]$ if both $x(t) and\ y(t)$ are _____ on $[a, b]$	Differentiable
267.	A curve $c: z(t) = x(t) + iy(t)$ for $a \leq t \leq b$, is said to be smooth curve if ________is continuous and nonzero on the interval.	$z'(t)$
268.	A curve C is said to ______ if it is constructed by joining finitely many smooth curves end to end.	Contour

269.	If $f(t) = u(t) + iv(t)$, where $u(t)$ and $v(t)$ are real valued functions of the real variable t for $a \le t \le b$, then $\int_a^b f(t)dt =$	$\int_a^b u(t)dt + i\int_a^b v(t)dt$
270.	For $f(t) = u(t) + iv(t)$, and $g(t) = p(t) + iq(t)$,we have $\int_a^b (f(t) + g(t))dt =$	$\int_a^b f(t)dt + \int_a^b g(t)dt$
271.	For $f(t) = u(t) + iv(t)$, and a partition of the interval $[a, b]$ given as $a \le t \le d$ and $d \le t \le b$,we have $\int_a^b (f(t) + g(t))dt=$	$\int_a^d f(t)dt + \int_d^b f(t)dt$
272.	Mean value theorem for real integrals: If $h(x)$ is continuous of $[a, b]$ then there exist $c \in [a, b]$ such that $\int_a^b h(x)dx =$	$h(x)(b - a)$
273.	Mean value theorem for real integrals does not hold in case of integrals of complex functions over ______ intervals.	Real
274.	Consider $f(t) = e^{it}$ for $0 \le t \le 2\pi$ then evaluate $\int_0^{2\pi} e^{it}\, dt =$	0
275.	For any $c \in [0,2\pi]$, we have $\lvert f(c)(2\pi - 0)\rvert = \lvert e^{it}\rvert \lvert 2\pi\rvert =$	2π
276.	A curve in the plane is defined as $c: z(t) = x(t) + iy(t)$ for $a \le t \le b$, where t is a ________	Parameter
277.	Evaluate the integral $\int_c \exp(z)\, dz =$	$\exp(a + ib) - 1$
278.	Evaluate the integral $\int_{c1} z\, dz$ and $\int_{c2} z\, dz =$	0
279.	Evaluate the integral $\int_c z\, dz =$ ______ where the contour C is circle from $-i$ to i. $c: z(t) = e^{it}$ for $-\frac{\pi}{2} \le t \le \frac{\pi}{2}$ =____	0, 0
280.	Evaluate the integral $\int_c z^a\, dz =$ _______. Where the contour C is semicircle from 2 to -2.	$\frac{2^{a+1}}{a+1}[(-1)^{a+1} - 1]$
281.	For a piece wise continuous complex valued function of real variable t , $f(t) = u(t) + iv(t); a \le t \le b$ we have $\left\lvert \int_a^b f(t)\, dt \right\rvert \le \int_a^b \lvert f(t)\rvert dt$	
282.	Each simple closed curve C divides the plane into two domains (Connected open set). One is bounded and is called ______ of C the other is unbounded and is called the exterior of C.	Interior
283.	Each simple closed curve C divides the plane into two domains (Connected open set). One is bounded and is called interior of C the other is unbounded and is called __________ the of C.	Exterior
284.	Simply connected domain: A domain D (Connected open set) is called _______connected domain if interior of every simple closed contour C is contained in D.	Simply
285.	A domain D that is not simply connected is called ________connected.	Multiply
286.	Positive Direction: ___________direction of a contour C is the direction corresponding to increasing values of a parameter t.	Positive

287.	Negative Direction: ___________direction of a contour C is the direction opposite to the positive direction.	Negative
288.	The line integral $\int_c ydx + xdy =$ ___________	0
289.	Jordon Curve Theorem: Each simple closed curve C divides the plane into two domains (Connected open set). One is bounded and is called interior of C the other is unbounded and is called the exterior of C.	
290.	In case of simple closed contour positive orientation (Contour Clockwise direction) and negative direction corresponds to clock wise direction, if C is positively oriented then –C is negatively oriented.	
291.	**Green's theorem:** Let C be a simple closed contour with positive orientation and Let R be the domain that forms the interior of C. If P and Q are continuous and have continuous partial derivatives then $\int_c P(x,y)dx + Q(x,y)dx = \int_R\int Q_x(x,y) - P_y(x,y)dxdy$ is called green's theorem.	
292.	**Cauchy Theorem:** Let f be analytic in simply connected domain D and f is continuous in domain D. If C is simple closed contour that lies in D then $\int_c f(z)dz = 0$	
293.	**Cauchy-Goursat Theorem:** Let f be analytic in simply connected domain D and f is continuous in domain D. If C is simple closed contour that lies in D then $\int_c f(z)dz = 0$	
294.	Let Z_0 *and* Z_1 be points in a domain D. A contour integral $\int_c f(z)dz$ is said to be independent of the path if its value is the same for all contour C in D with initial point Z_0 and terminal point Z_1	
295.	Generalization of Cauchy-Goursat Theorem: Let C_1 *and* C_2 be two simple closed positively oriented contours such that C_2 lies interior to C_1. If f(z) is analytic in a domain D that contains both C_1 and C_2 then $\int_{c1} f(z)dz = \int_{c2} f(z)dz$	
296.	Remember: $\int_{c1} \frac{dz}{(z-z0)^n} = 2ni \; if \; n = 1$ and $\int_{c1} \frac{dz}{(z-z0)^n} = 0 \; if \; n \neq 1$	
	MODULE#101 to121	
297.		
	FROM PAST PAPERS	
298.	The scalar multiple of a vector $\vec{x} = (x_1, x_2, x_3)$ by a real number k is the point $k\vec{x} = (kx_1, kx_2, kx_3)$, clearly _____ for all k and $\vec{x}$.	Both b and c
299.	Let $f(z)$ be analytic function and C is contour enclosing point x_0, then Cauchy Integral Formula states f(z0)=	$\frac{1}{2\pi i}\int_c \frac{f(z)}{z - z_0}dz$

300.	If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as Frenet formula (formulas)?	All a,b,c
301.	An important consequences of the wedge product is that "repeats are zero" due to "alternation rule", that means____	All a,b,c
302.	If the three vectors $\vec{t}, \vec{n}$ *and* $\vec{b}$ are perpendicular to each other and form the Frenet frame , then $\vec{b} \times \vec{t} =$ ____	$\vec{n}$
303.	If p *and* q are points in R^3 then __________ is the symmetric property of dot product.	$p.q = q.p$
304.	The argument of complex number $(1+i)^{\pi}$ is _____	90^0 (Not confirm)
305.	If $f(x) = 3x^2 - 2x + 4$ then $f'(x) =$______	$6x - 2$
306.	If $f(z) = z^3 - 3z + 12$ then $f'(z) =$______	$3z^2 - 3$
307.	Let $f(z) = \frac{1}{z^2+1}$ is discontinuous at z=______	$\pm i$
308.	If $u(x,y) = 5x(4-2y)$ is harmonic in some domain then harmonic conjugate v(x,y)=	$20y - 5y^2 + f(x)$
309.	Any polynomial $\mathrm{p(z) = a_0 + a_1z + a_2z^2 + \cdots \ldots \ldots . + a_nz^n (a_n \neq 0)}$ of degree $\mathrm{n(n \geq 1)}$ has least one zero, i.e. there exists at least -------------$\mathrm{Z_0}$ such that $\mathrm{P(Z_0) = 0}$	One point
310.	Any function which is analytic at a point x_0 must have a _______________ series about x_0.	Taylor's
311.	Harmonic conjugate of u(x,y)=e^y cos x is ________.	-e^y Sin x + C
312.	The point at which $f(z) = \frac{1}{z^2+1}$ is not analytic are;	*i and -i*
313.	In the trigonometric functions of complex numbers Cos x=	$\frac{e^{ix} + e^{-ix}}{2}$
314.	Let Z be a complex valued function then arcsin(Z)=	$-ilog(iz + (1-z^2)^{\frac{1}{2}})$
315.	Let C denote a contour of length L, and suppose that a function f(z) is piece wise continuous on C. If M is non negative constant such that $\lvert f(z) \rvert \leq M$ for all points z on C at which f(z) is defined, then _____	$\left\lvert \int_c f(z)dz \right\rvert \leq ML$
316.	The Cauchy-Goursat theorem shows that _____	$\int_c \mathrm{f(z)dz} = 0$
317.	The derivative of a complex valued function $w(t) = u(t) + iv(t)$ *is* ________, where u and v are real valued functions of a real variable t.	$w'(t) = u'(t) + iv'(t)$ *is*
318.	The circle $z = e^{i\theta}, 0 \leq \theta \leq \pi$ is centered at _____	(0,0)
319.	The principal value of i^i =	$exp(-\frac{\pi}{2})$